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Deformation behaviour of iron-rich iron-aluminum alloys at low temperatures

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Abstract

The deformation behaviour of binary monocrystalline and polycrystalline Fe-Al alloys with Al contents up to 18 at.% and only low unavoidable impurity contents—in particular less than 100 wt.ppm C—has been studied at room temperature and −100 °C. The effects of quenching and annealing treatments on the behaviour of as-cast materials were investigated in order to clarify the dependence of strength and ductility on Al content and short-range ordering. It was found that the stress-strain behaviour at low temperatures is controlled primarily by Al solid-solution hardening and quenched-in excess vacancies with only minor effects of short-range ordering.
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Keywords: Metal & Alloys (iron alloys); Ordering (short range); Mechanical properties (yield phenomena)

1. Introduction

The alloying of iron with aluminium produces solid solution hardening [1,2], reduces the density and provides a high oxidation and high-temperature corrosion resistance [3–5]. Thus Fe-Al alloys are attractive for structural applications. A problem is posed by the decrease of ductility with increasing Al content [1–3,6,7]. This problem is aggravated by ordering reactions, which occur at higher Al contents beginning with about 10 at.% Al and which affect strength and ductility [8,9] as well as the elastic behaviour [10,11]. At such high Al contents there is a transition from the disordered atom distribution to an intermediate state, which is known as K-state, and then to long-range ordering to produce the DO_3 crystal structure and the B2 crystal structure with increasing Al content [12]. Apart from ordering, Al segregation at grain boundaries may contribute to embrittlement [13]. It has to be noted that these various studies referred to Fe-Al alloys which contained comparatively high amounts of carbon in the range of 300–600 wt.ppm as impurities.

In view of possible automotive applications, a major cooperative research project was initiated to explore the possibilities for developing Fe-Al materials with high strength, low density and sufficient ductility for producing sheet material [14,15]. Within this project work was directed at

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clarifying the mechanical behaviour of Fe-Al alloys with Al contents in the intermediate K-state range. The K-state, which is characterised by complex short-range ordering reactions of still unclear character, is subject of a separate study [16]. The present study is focussed on the mechanical behaviour of binary Fe-Al alloys at room temperature and −100 °C. The deformation behaviour at higher temperatures as well as the effects of ternary alloying additions are subject of separate reports [17,18]. Preliminary results have been presented previously [19]. Further details are available in ref. [20].

2. Experimental

2.1. Characterisation of alloys

Binary Fe-Al alloys with Al contents in the range 4–18% Al (always at.% if not stated otherwise) were prepared by vacuum induction melting in Al_2O_3 crucibles using Fe with 99.99% purity (purified by zone melting to minimise the impurity content) and Al with 99.999% purity and solidification in Cu moulds to obtain rods usually with 18 mm diameter. Levitation melting was used only for selected alloys.

Monocrystals with up to 10 cm length were grown by the Bridgman method (10–15 mm/h pulling rate). The crystal growth with subsequent cooling occurred within a time period of about 12 h. The monocrystals were etched by ammonium persulfate ($(NH_4)_2S_2O_8$) and oriented using a special laser light-figure method [21]. The alloy compositions were usually determined by inductively coupled plasma (ICP) analysis. All alloys studied are listed in Table 1 with their compositions and possible impurity contents.

Specimens were cut by electrostatic discharge machining, mechanically polished and cleansed by ultrasound in acetone (except for the monocrystals). The specimens were heat-treated at temperatures up to 600 °C usually for less than 14 days, at higher temperatures less than 4 days and at 1100 °C usually only 15 min. The short heat treatments at temperatures up to 300 °C occurred in an argon atmosphere with 99.99% purity. For all other heat treatments the specimens were enclosed in evacuated (10^{-6} bar) SiO_2 capsules. Any uptake of Si by the possible reaction of the specimens with the SiO_2 capsules could not be detected for the chosen heat-treatment conditions.

The specimens were etched by an alcoholic 10% nitric acid solution for metallographic inspection by optical microscopy for revealing grain boundaries and precipitates. Scanning electron microscopy (SEM) was used for studying fracture surfaces and for qualitative precipitate analysis (by energy-dispersive X-ray diffraction (EDX)). For transmission electron microscopy (TEM) using a Philips CM20 instrument with 200 kV thin foils were prepared by electrolytic twin jet polishing at –30 °C and 12 V voltage using a 7:3 mixture of methanole and nitric acid. In addition, ion milling was applied if necessary.

2.2. Mechanical testing

For hardness testing the microhardness tester Fischerscope H100 (with maximum load of 1 N) was used which allows the determination of the elastic and plastic deformation [22]. The respective so-called universal hardness *HU* was determined as $HU = F/(26.43 \cdot h^2)$ [N/mm^2] with the indentation force F and the indentation depth h. The reported *HU* values are averages of more than 40 measurements.

The yield stress was determined as 0.2% proof stress in compression (with total straining of at least 5%) at room temperature and −100 °C with rates of 10^{-4} s^{-1} and 10^{-2} s^{-1}. The surfaces of the specimens with dimensions $5 \times 5 \times 10$ mm^3 were cleansed by grinding and the monocrystals were polished with 3 μm diamond paste before testing. Ductility was studied by tensile testing at rates between 10^{-4} s^{-1} and 10^{-2} s^{-1}. For this, cylindrical specimens were prepared by turning according to German standard DIN 50125.

3. Results

3.1. Microstructure

The polycrystalline alloys (Table 1) with Al contents in the range 4–18% Al (always at.% if not

Table 1
Chemical compositions of alloys studied

Al content			Impurities		
Nominal	Real		C	Si	other[a]
at.%	at.%	wt.-%	wt. ppm		
Monocrystals:					
16	16.0	8.4	45	40	O: 60
17	16.8	8.9	70	140	Cu: 50 Mo: 90
Polycrystals:					
0	n.d.	n.d.	100		
4	4.1	2.0	60	100	Ni: <40
6	6.0	3.0	50	70	
7	n.d.	n.d.	n.d.		
8	7.9	4.0	50	80	
9	9.1	4.6	60	80	
10	n.d.	n.d.	n.d.		
11	10.9	5.6	60	80	
12	12.0	6.2	50–60	60–90	Cu: <40
13	12.9	6.7	70–80	70	As: <30
14	13.8–14.0	7.2–7.3	20–70	50–60	Ni: <40 Cu: <30
15	14.9	7.8	20	60	Ni: <40
16	16.0–16.1	8.4–8.5	<20–40	<20–80	Co: <30 Ni: <40 As: <30
17	16.6–17.0	8.8–9.0	<20–70	60–110	As: <50 Ni: <90
18	17.5–18.0	9.3–9.6	40–70	<20–100	As: <70 Ni: <40

n.d.: not determined.
[a] if >20 ppm As, Ca, Co, Cu, Cr, Mg, Mn, Mo, Nb, Ni, P, Si, Sn, Ti, V, or >10 ppm N, O, S.

stated otherwise) showed a coarse grain structure with grain sizes of the order of 1 mm. The as-cast alloys contained no second phases. Fine thin plate-like or rod-like precipitates were revealed by metallographic inspection on grain boundaries after slow furnace cooling from 1100 °C or prolonged heat treatments of 14 days at 320 °C, which are believed to be carbides due to the presence of up to 80 ppm carbon (always wt.ppm). Indeed the amount of these precipitated particles was smaller for lower C contents and only rare tiny precipitate particles were detected on grain boundaries of an alloy with only 20 ppm C after furnace cooling.

The monocrystals contained rare globular AlN inclusions with diameters up to 30 μm which were identified by EDX analysis. The thin carbides, which were observed only on grain boundaries in the polycrystals, were found in the Fe-17%Al monocrystal with 70 ppm C after a heat treatment of 14 days at 320 °C with concurrent 100 MPa loading in <111> direction. Without loading much less carbides were observed. No carbides were detected in the Fe-16%Al monocrystal with 45 ppm C after a heat treatment of 14 days at 320 °C. Without heat treatment no carbides were found in all monocrystals.

3.2. Hardness

In view of possible effects of atomic order on the mechanical behaviour, specimens with (111)

orientation of the Fe-16% monocrystals (Table 1), which were studied with respect to possible ordering reactions [16,20], were subjected to various heat treatments for affecting the state of atomic order. The obtained microhardness *HU* data are shown in Fig. 1. Obviously the observed hardness does not depend sensitively on heat treatment since the data vary only by about 10% at most. In particular, the data for complete furnace cooling (1100 °C FC RT in Fig. 1) and interrupted furnace cooling with subsequent quench (1100 °C FC 600 °CQ in Fig. 1) or quench from 1100 °C (Q∗) and quench with subsequent annealing do not differ significantly.

3.3. Strength and ductility

Fig. 2 illustrates the compressive stress-strain behaviour of the above Fe-16% monocrystal at room temperature as a function of prior heat treatments. As in the case of hardness, the data indicate only small effects of heat treatments on the deformation behaviour.

The variation of the yield stress with Al content was studied in compression using polycrystalline alloys (Table 1). Usually two specimens of each alloy were tested to obtain yield stress data which differ by 3% at most. The yield stress for pure iron was obtained by averaging the data for 3 rather coarse-grained specimens (with perpendicular orientations in the original ingot) of iron which was prepared by electron-beam remelting. The results are shown in Fig. 3. Obviously there is a linear increase of the yield stress with increasing Al con-

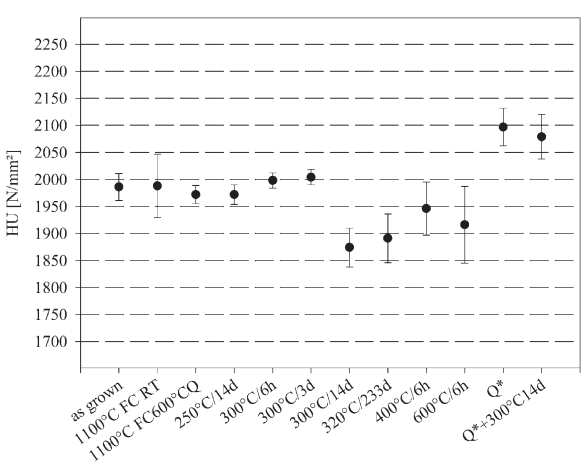


Fig. 1. Microhardness *HU* at room temperature of (111) oriented Fe-16%Al monocrystals as a function of prior heat treatment (1100 °C FC RT: furnace cooling from 1100 °C to room temperature; 1100 °C FC 600 °CQ: furnace cooling from 1100 °C down to 600 °C with subsequent water quench; Q∗: water quench from 1100 °C).

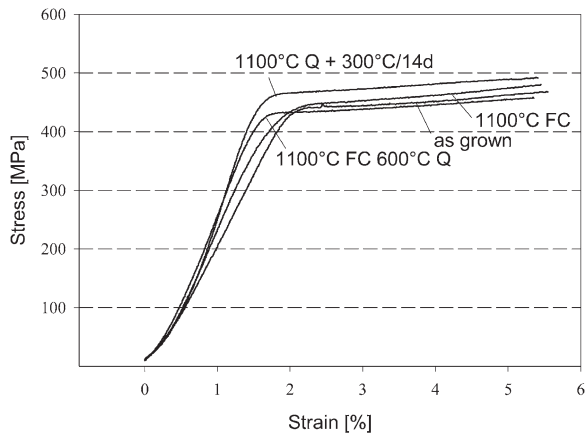


Fig. 2. Compressive stress-strain curves (10^{-4} s^{-1} compression rate) at room temperature of [112] oriented Fe-16%Al monocrystals as a function of prior heat treatment (1100 °C FC: furnace cooling from 1100 °C to room temperature; 1100 °C FC 600 °CQ: furnace cooling from 1100 °C down to 600 °C with subsequent water quench; 1100 °CQ + 300 °C/14: water quench from 1100 °C with subsequent annealing at 300 °C for 14 days).

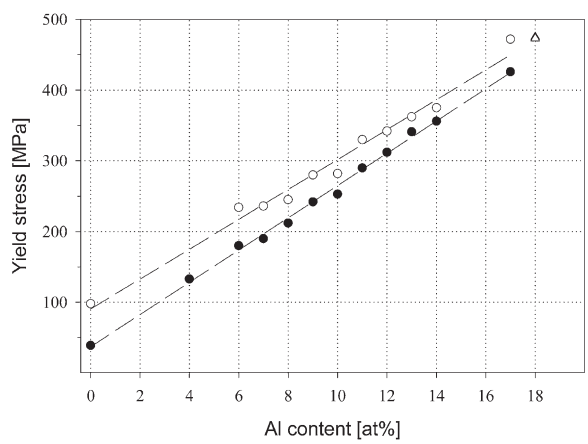


Fig. 3. Compressive yield stress at 10^{-2} s^{-1} (○) and 10^{-4} s^{-1} (●) compression rate as a function of Al content for polycrystalline as-cast Fe-Al alloys at room temperature. The Fe-18%Al alloy (Δ) yielded with concurrent twinning at both rates.

tent at both compression rates at room temperature. The interpolating straight lines indicate a yield stress increase of 21and 23 MPa per unit% for the higher rate and the lower rate, respectively. Deformation twinning was observed for the Fe-18%Al alloy at both rates.

The variation of ductility with Al content was studied in tension using polycrystalline alloys of Table 1. The observed stress-strain behaviour at room temperature and –100 °C is exemplified by the curves in Fig. 4. Clearly an increased Al content not only increases the strength, which corresponds to the observed compressive yield stress behaviour, but also reduces the fracture strain. The room temperature curves obviously indicate ductile fracture whereas the curves at –100 °C indicate an increasingly brittle fracture with increasing Al content. The coarse serrations of the Fe-18%Al curve are due to deformation twinning as was revealed by metallographic observations. The most ductile Fe-4%Al shows a yield stress drop at –100 °C, which was also observed for Fe-6%Al.

The results of the tension tests are summarised in Fig. 5 (the data were obtained by averaging 3–4 tests; the respective yield stress and strength data differed by 3% at most). The yield stress in tension equals that in compression (Fig. 3) referring to the same rate and temperature. The ultimate tensile strength parallels the yield stress. At the lower temperature of −100 °C the yield stress and ultimate tensile strength values are higher than those at

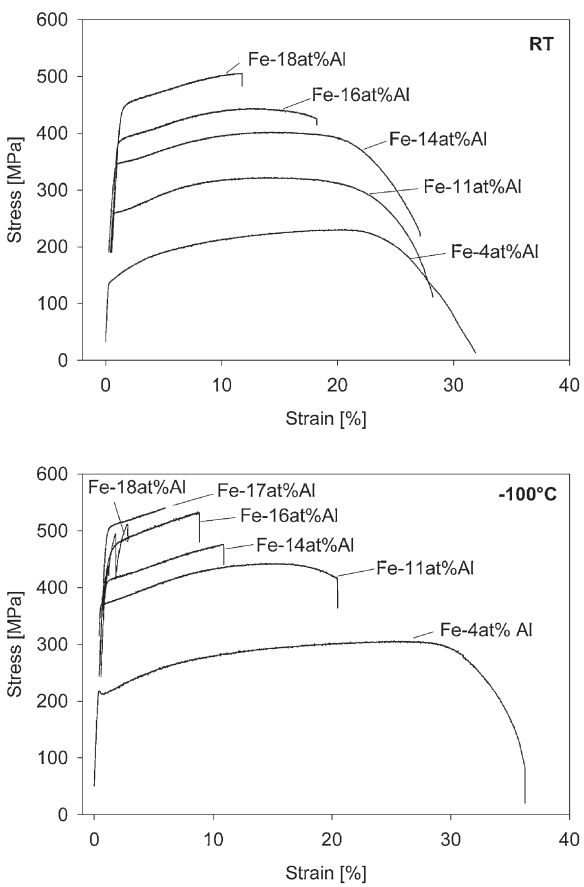


Fig. 4. Tensile stress-strain curves (10^{-4} s^{-1} tension rate) for various polycrystalline as-cast Fe-Al alloys of Table 1 at room temperature RT (a) and −100 °C (b).

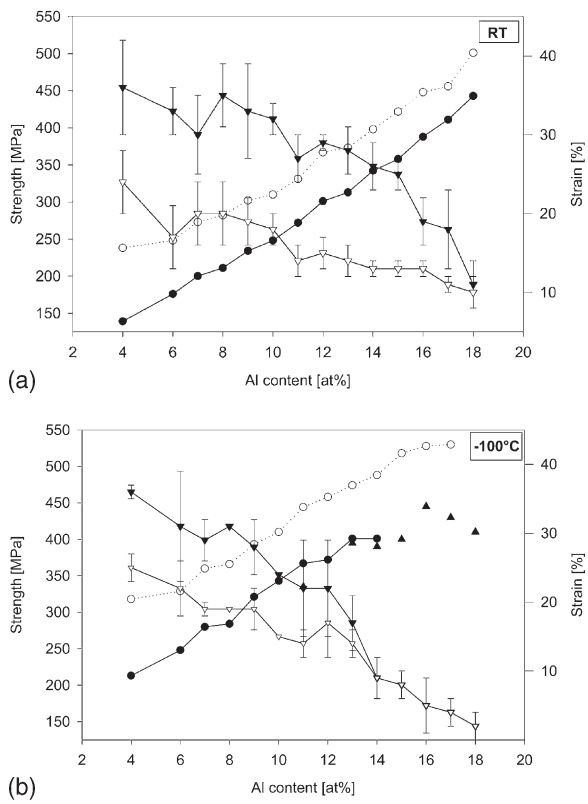


Fig. 5. Tensile yield stress (●), ultimate tensile strength (○), strain before necking (▽) and fracture strain (▼) at 10^{-4} s^{-1} tension rate for various polycrystalline as-cast Fe-Al alloys of Table 1 as a function of Al content at room temperature RT (a) and −100 °C (b); in addition the stress for initiating deformation twinning (▲) at −100 °C is plotted in Fig. 5(b) as a function of Al content.

room temperature by about 80 MPa. Deformation twinning was observed at the lower temperature for higher Al contents, i.e. occasionally for 11–14% Al and above 14% Al in all specimens. The respective stresses for initiating deformation twinning are shown in Fig. 5(b).

The strain before necking in Fig. 5 decreases with increasing Al content. Likewise the higher fracture strain decreases with increasing Al content, however, with a steep decline above 15% Al at room temperature and above 12% Al at −100 °C to reach the strain before necking. Obviously the fracture without prior necking, i.e. brittle fracture, occurs at room temperature for an Al content of 18% and at −100 °C for Al contents above 13%. Both strain before necking and fracture strain show apparent relative maxima at about 8 and 12% at room temperature, which is less pronounced at −100 °C.

The dislocation distribution after deformation was studied by transmission electron microscopy (TEM). In the case of the Fe-12%Al alloy (with 60 ppm C), which showed only a weak effect of quenching, glide bands were observed with dislocation tangles between them after deformation of the quenched alloy (Fig. 6(a)) whereas the deformation of the furnace-cooled alloy initiated cell formation (Fig. 6(b)). Before deformation only few straight dislocations were observed in the quenched alloy whereas nearly no dislocations were observed in the furnace-cooled alloy. In the case of the Fe-16%Al alloy (with less than 20 ppm C), which showed a strong effect of quenching, glide bands were observed in various directions with primarily straight dislocations between them after deformation of the quenched alloy (Fig. 7(a)) whereas the deformation of the furnace-cooled alloy lead to dislocation tangles between the glide bands (Fig. 7(b)).

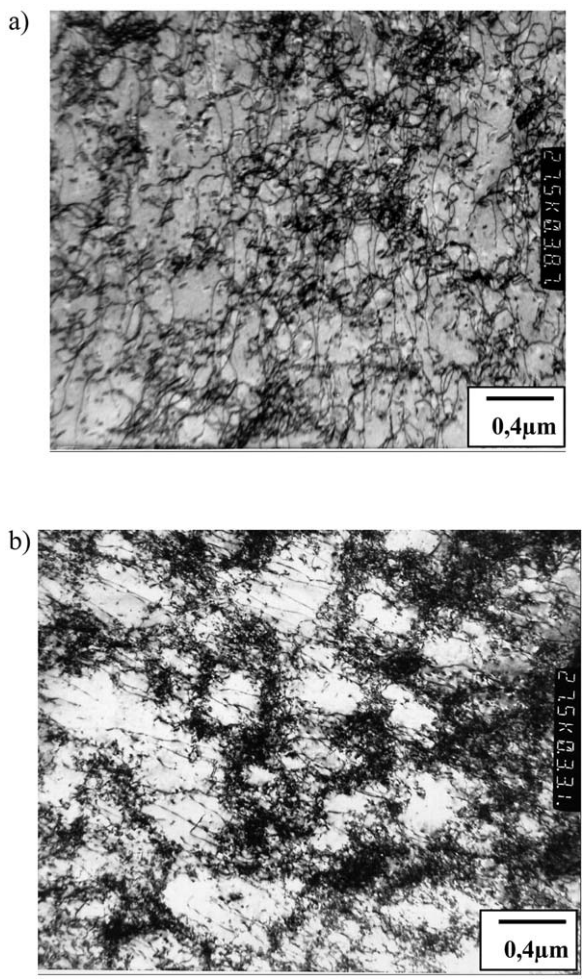


Fig. 6. Transmission-electron micrograph of the dislocation distributions in polycrystalline Fe-12%Al (with 60 ppm C) with quench (a) and furnace cool (b) from 1100 °C with subsequent 5.5% compressive straining at 10^{-4} s^{-1} rate at room temperature.

4. Discussion

4.1. Effects of ordering

The occurrence of ordering reactions in Fe-Al alloys was studied separately [16,20]. It was found that noticeable short-range ordering occurs in Fe-Al alloys with Al contents in the range 13–18% (always at%) at temperatures below 600 °C during cooling after solidification. The highest degree of short-range ordering (as characterised by x-ray scattering intensity) was observed at 250 °C (for isochronal annealings of Fe-18%Al for 14 days). The activation energy for this ordering reaction was found as only 0.3 eV/atom and indeed 100 h were sufficient at 250 °C for reaching a near-equilibrium state. Quenching from 1100 °C suppressed short-range ordering nearly completely.

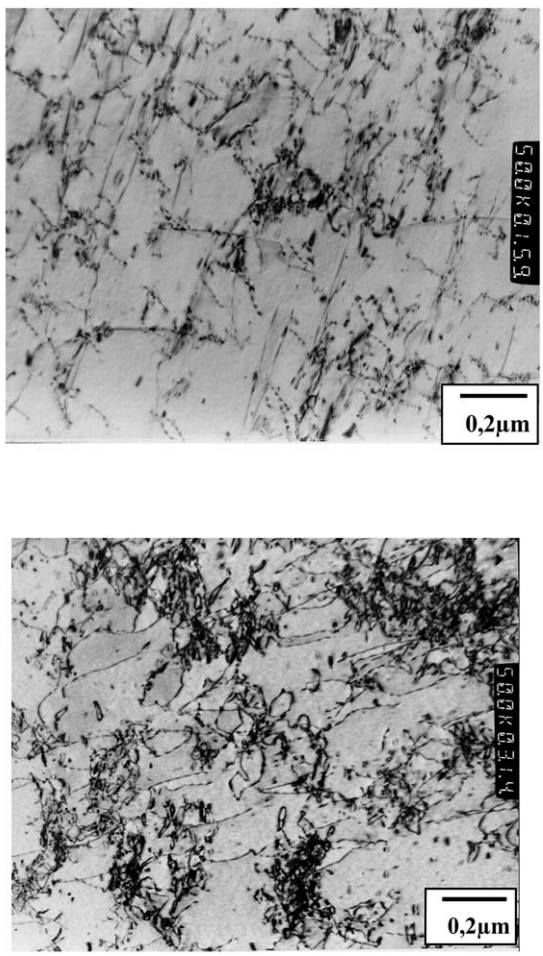


Fig. 7. Transmission-electron micrograph of the dislocation distributions in polycrystalline Fe-16%Al (with less than 20 ppm C) with quench (a) and furnace cool (b) from 1100 °C with subsequent 5.5% compressive straining at 10^{-4} s^{-1} rate at room temperature.

In view of the above findings the various heat treatments of the Fe-16%Al monocrystal specimens for the hardness tests in Fig. 1 result in different states of short-range order, i.e. maximum ordering after the 250 °C/14 day anneal (at 250 °C for 14 days), minimum ordering after quenching from 1100 or 600 °C and intermediate degrees of order for the other treatments in Fig. 1. However, the hardness results do not reflect this pattern. Both the 250 °C/14 day anneal for maximum ordering and the quench from 600 °C (after furnace cooling from 1100 °C down to 600 °C) for minimum ordering as well as furnace cooling from 1100 °C to room temperature and anneals at 300 °C for 6 h and 3 day with intermediate ordering produce the same hardness which equals that of the as-grown monocrystal. Lower hardness values were obtained by the longer anneal at 300 °C for 14 days and the anneals at higher temperatures whereas higher hardness values were obtained by quenching from 1100 °C without and with subsequent anneal. It is concluded that short-range ordering has no major impact on hardness.

The stress-strain curves in Fig. 2 show a similar behaviour. The curves for the as-grown and furnace-cooled materials differ only little from the material with furnace cooling from 1100 °C down to 600 °C and subsequent quench whereas a higher yield stress was produced by quenching from 1100 °C and subsequent anneal. It is concluded that only heat treatments at temperatures above 600 °C affect the mechanical behaviour of Fe-Al alloys significantly.

4.2. Effects of quenched-in excess vacancies

It is well known that the concentration of thermal vacancies increases with increasing temperature, i.e. annealing at high temperature produces additional vacancies which are trapped as excess vacancies by quenching to lower temperatures. The vacancy formation enthalpy decreases with increasing Al content for Fe-Al alloys with Al contents in the range 7–30% and is about 1–1.2 eV for Fe-Al with 16–18% Al [23,24]. This comparatively low value makes vacancy formation easy. Consequently a high concentration of excess vacancies is expected in the studied Fe-Al alloys that are quenched from 1100 °C, which increases with increasing Al content. Excess vacancies at low temperatures are immobile, i.e. they are effective obstacles to dislocation movement and contribute to hardening as was reported in particular for B2 ordered Fe-Al alloys with higher Al contents [25,26]. Indeed the highest hardness is shown in Fig. 1 by the specimen which was quenched from 1100 °C to room temperature.

Similar effects have been observed for Fe-Al alloys with much higher Al-content, i.e. for Fe-40%Al which is an ordered intermetallic phase

with B2 structure [27]. Both the hardness and the yield stress were found to increase steeply with increasing quench temperature for quench temperatures above 500 °C. This was attributed to vacancy hardening, i.e. pinning of dislocations by single vacancies. Indeed a linear relationship between the microhardness and the square root of the vacancy concentration was observed for such alloys with 40–50%Al [28]. It was further found that a high vacancy concentration increases the fracture strength and decreases the elongation which was attributed to vacancies promoting the fracture along slip planes [25].

Excess vacancies get healed out by annealings below the quench temperature. In the case of Fe-Al alloys with Al contents up to 20% excess vacancies were reported to heal out at temperatures between 300 and 400 °C [29]. Indeed the hardness of the quenched specimen in Fig. 1 is slightly reduced by the anneal at 300 °C after quenching. However, these effects are only small in view of the data for the specimens without quench. This may indicate rather slow kinetics of the healing out of excess vacancies. The kinetics are controlled by the vacancy migration enthalpy which is comparatively high for Fe-Al alloys with higher Al contents [30–35]. This migration enthalpy decreases only with increasing temperature and reaches a minimum for Fe-25%Al [30]. However, the reported data are still conflicting with 1.3 eV for pure bcc iron [34] and 0.5–1.6 eV for Fe-25%Al [33]. Thus it can only be concluded that these vacancy data are not in contrast to the observed effects of anneals on the behaviour of quenched Fe-Al alloys. Indeed the equilibration of the vacancy concentration was reported to be very slow and complete healing out was reached only by cyclic annealing treatments [36]. This would mean that the annealing at 300 °C of the alloys in Fig. 1 even for 14 days may not have been sufficient for the complete elimination of the excess vacancies.

It is noted that the present results are in contrast to the findings of Davies, who reported a 25% increase of the yield stress of a quenched Fe-16.5%Al alloy by a short 1 h anneal at 300 °C without, however, giving information on alloy preparation and impurity content [8].

4.3. Effects of Al content

Both Fig. 3 and Fig. 5(a) indicate a linear increase of the yield stress with increasing Al content for alloys with Al contents up to 18%. This fits Suzuki's theory of solid-solution hardening for bcc alloys [37] which has been found to describe the solid-solution hardening of various binary bcc Fe-base alloys quite well [38]. In particular, a linear increase of the yield stress with the solute content was found for Fe-Al monocrystals (single-slip orientation) with 2–6%Al already in the past [39]. However, the reported hardening coefficient of the order of 1 GPa contrasts with the present value of about 2.2 GPa in Figs 3 and 5(a), (b) for polycrystalline alloys with Al contents in the range 0–18%, which is not sensitive to deformation rate (Fig. 3) and temperature (Fig. 5). This contrast is believed to be due to differences in alloy preparation (monocrystals with annealing at 900 °C for 24 h and subsequent air cooling vs. as-cast polycrystals) and differences in slip (single [111](110) slip in monocrystals vs. multiple slip in polycrystals). Earlier data by Morgand et al. for polycrystalline alloys (with C contents up to 120 ppm) also show a linear increase of the yield stress with increasing Al content with a hardening coefficient of about 1.8 GPa at 100 °C for Al contents up to 18% [2]. It is concluded that the presently observed linear increase of the yield stress with increasing Al content is produced exclusively by solid solution hardening which is described by Suzuki's theory. The deviations from the linear behaviour at −100 °C, which are visible in Fig. 5(b), are attributed to additional deformation twinning.

The decrease of ductility as characterised by the decrease of fracture strain in Fig. 5(a), (b) with increasing Al content is less simple, i.e. there is a linear decrease with increasing Al content at both temperatures only for the elongation before necking which is overlayered by few small deviations to higher and lower strains. The fracture strain parallels the elongation before necking with similar deviations up to about 15%Al at room temperature and to about 12%Al at −100 °C. At higher Al contents the fracture strain decreases with increasing Al content more steeply to reach the elongation before necking at 18%Al at room temperature and

at 14%Al at −100 °C. Obviously there is a transition from ductile fracture with necking to brittle fracture without necking for Al contents in the range of 15–18%Al at room temperature and of 12–14%Al at −100 °C. It is noted that a similar transition was reported for Fe-Al alloys again with 15–19%Al, but with much higher carbon contents of about 300–500 ppm [1].

The TEM observations (Figs 6–7) reveal a reduced dislocation mobility for the higher Al content of 16% Al since distinct cell formation, which requires a sufficient dislocation mobility, was observed after deformation only in the furnace-cooled Fe-12%Al alloy. The latter observation means that quenching before deformation also reduces the dislocation mobility which is believed to be due to the presence of excess vacancies—see preceding section. A reduced mobility of screw dislocations is a prerequisite for the formation of twins in bcc crystals and this may indeed be achieved by substitutional alloying [40]. It is concluded that the decrease of the fracture strain with increasing Al content results from the decreasing dislocation mobility which decreases with increasing Al content, and this is enhanced by quenched-in excess vacancies.

4.4. Yield stress drop and stress-strain serrations

Yield stress drops were produced only by Fe-4%Al (Fig. 4(b)) and Fe-6%Al at −100 °C. The few previous studies of the deformation behaviour of disordered Fe-Al alloys with Al contents below 20% did not mention the presence or absence of yield stress drops and stress-strain serrations [1,2]. However, such phenomena were reported repeatedly and studied in detail for Fe-Al alloys with higher Al contents above 20%, i.e. ordered alloys with $D0_3$ or B2 structure depending on Al content and temperature [41–47]. Careful strain-ageing experiments with Fe-40%Al alloys led to the conclusion that yield drops are shown at low temperatures only by B2-ordered Fe-Al alloys and the effect decreases with decreasing Al content [48]. These yield drop effects were attributed to specific dislocation configurations in the B2 structure and are not related to grain size or quenched-in vacancies. However, a study of B2-ordered Fe-Al monocrystals did not reveal pronounced yield stress drops [47]. Likewise $D0_3$-ordered Fe-Al monocrystals with 25–30% Al did not show yield stress drops at low temperatures [49,50]. Thus the various studies of the low-temperature deformation behaviour of the ordered Fe-Al alloys with Al contents above 20%Al do not offer any possible explanations for the observed low-temperature yield stress drops of the alloys with only 4 or 6%Al.

Fine stress-strain serrations at room temperature were previously reported for B2-ordered fast-cooled monocrystals with 33–34%Al [47]. These serrations were found to be produced by quenched-in excess vacancies, which contributed distinctly to strengthening and reduced ductility. In Section 4.2. it was found for the studied alloys that quenching produces excess vacancies, which contribute significantly to strengthening and embrittlement and heal out only slowly. It is concluded that the yield stress drops, which were observed at −100 °C for the as-cast alloys with low Al contents, may be related to excess vacancies. However, more detailed studies are necessary for clarifying this and providing clear evidences. Finally it is noted that the possibility of deformation by formation of micro-twins with barely visible stress-strain serrations was discussed with respect to ordered Fe-23%Al at low temperatures [51]. However, no evidence was found for this for the present alloys.

In addition, deformation twinning occurs as is indicated by the large load drops in Fig. 4(b) with audible clicks and by metallographic evidence (Figs 3, 4(b) and 5(b)). Deformation twinning with large load drops at low temperatures has often been observed from the beginning of deformation in many bcc metals and alloys [52]. This is due to an insufficient number of active dislocation glide systems thus depending sensitively on crystal orientation and is enhanced by decreasing the temperature. In particular, a reduced mobility of screw dislocations is a prerequisite for the formation of twins in bcc crystals and this may indeed be achieved by substitutional alloying [40]. In the present case of Fe-Al alloys, deformation twinning was found at room temperature only for the Fe-18%Al alloy (Fig. 3), which agrees with early observations [51,53]. At −100 °C deformation twinning occurred already for lower Al contents in the range

of 11–18%. It is again concluded that the number of sufficiently mobile dislocations decreases with increasing Al content (as was already discussed in the preceding Section 4.3.) and decreasing temperature.

5. Conclusions

The deformation behaviour of variously treated binary Fe-Al alloys with Al contents up to 18 at.% and only low unavoidable impurity contents—in particular less than 100 wt.ppm C—has been studied at room temperature and −100 °C. The following conclusions are drawn from the results.

- Short-range ordering in the studied alloys, which is controlled by heat treatments at temperatures below 600 °C, has no major impact on the deformation behaviour. Only heat treatments at temperatures above 600 °C affect the mechanical behaviour of Fe-Al alloys significantly.
- Quenching from high temperatures produces excess vacancies which contribute to hardening. Softening by subsequent anneals to eliminate the excess vacancies is a slow process.
- The yield stress of the studied alloys at room temperature increases linearly with increasing Al content up to 18 at.% Al which corresponds to Suzuki's theory of solid-solution hardening for bcc alloys. This composition range includes alloys with short-range order.
- Corresponding to the increasing yield stress by solid-solution hardening, the ductility as characterised by elongation before necking and fracture strain decreases with increasing Al content with ductile fracture for Al contents up to about 15 at.% Al at room temperature and to about 12 at.% Al at −100 °C and a transition to brittle fracture without necking in the range of 15–18 at.% Al at room temperature and of 12–14 at.% Al at −100 °C.
- The yield stress drops at low temperatures are attributed to the interaction of mobile dislocations and vacancies.
- Deformation twinning at low temperatures occurs because of insufficient dislocation mobility the more readily the lower the temperature and the higher the Al content is.

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